

Worksheet — Laws of Motion (NIOs Official)

Total marks: **30** · Questions: **10**

Instructions: This is the NIOS Senior Secondary Physics (312) official Worksheet-3 for Lesson 3. Attempt all questions. Most are open-ended and inquiry-based — observe your surroundings, identify forces, and explain in your own words. Show your steps for numerical questions. (Take $g = 10 \text{ m/s}^2$ unless otherwise stated.)

Q1. Although a force is not visible, its effect can be seen or experienced. Observe your surroundings and mention different kinds of EFFECTS of forces on different objects. [3 marks]

Q2. Why do passengers standing in a bus fall in the BACKWARD direction when the stationary bus begins to move suddenly, and pushed FORWARD when a moving bus stops suddenly? [3 marks]

Q3. A 200-kg stone is moving toward you at 2 m/s. At the same time, a 2-kg bullet at 100 m/s is also moving toward you. Explain with reasons which one is more dangerous and requires more force to stop. [3 marks]

Q4. Apply Newton's third law of motion to explain the movements of a ball during a game of football. [3 marks]

Q5. Newton's First Law (Law of Inertia) states that 'a body continues to be in a state of rest or of uniform motion in a straight line unless it is acted upon by a net external force.' Now roll a ball along the ground and observe its motion. Write your observation and explain why the ball comes to rest after travelling some distance even though no external force seems to act on it. [3 marks]

Q6. Observe your surroundings and give TWO examples of friction that are useful and TWO examples of friction that are not useful. Explain how you would increase friction where it is useful and reduce friction where it is not. [3 marks]

Q7. Which diagram represents balanced forces? How do opposing forces affect the motion of an object if the forces are balanced? [2 marks]

Q8. Take a solid object from your surroundings and measure its mass. Lift the load with the help of a rope at a constant acceleration to the roof of your house. Measure the height of the roof. Let the load cover this height in 1 minute. Calculate the tension in the rope. [4 marks]

Q9. Suppose you are standing on a weighing scale in a lift. If the lift is going DOWN with a downward acceleration of 9 m/s^2 , what would be the reading of the weighing scale? Justify your answer with explanation. (Take $g = 10 \text{ m/s}^2$ and your mass = 60 kg.) [3 marks]

Q10. It is generally observed that roads on hilly areas or mountain roads are winding upwards rather than going straight up. Give reasons. [3 marks]

Answer Key

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Q1. Although a force is not visible, its effect can be seen or experienced. Observe your surroundings and mention different kinds of EFFECTS of forces on different objects.

Answer: Effects of force (with examples): (1) Change in state of motion — a kicked football accelerates from rest; brakes stop a moving car. (2) Change in direction of motion — wind deflects a flying kite; a batsman changes the direction of a cricket ball. (3) Change in shape — squeezing a sponge or rubber ball; clay being moulded. (4) Change in size — stretching a rubber band; compressing a spring. (5) Rotation about an axis — pushing on a door rotates it about the hinges; turning a steering wheel. (6) Change in speed without changing direction — pushing harder on a stationary cycle to make it move. All these are observable manifestations of forces.

Look for at least four valid effects: change in motion (start, stop, speed), change in direction, change in shape/size, rotation.

SHORT_ANSWER · EASY · 12-NIOS-PHY-LM-01

Q2. Why do passengers standing in a bus fall in the BACKWARD direction when the stationary bus begins to move suddenly, and pushed FORWARD when a moving bus stops suddenly?

Answer: Both phenomena are consequences of inertia (Newton's First Law). (a) Bus starts suddenly: when the bus is at rest, passengers and the bus are both at rest. When the bus accelerates forward, the bus carries the passenger's feet forward (through floor friction), but the upper body, by inertia of REST, tends to stay where it was. The result: the upper body lags behind the feet, so the passenger appears to fall backward. (b) Moving bus stops suddenly: when the bus is moving, the passenger is moving forward with it. When the bus brakes, the bus and the passenger's feet decelerate quickly, but the upper body, by inertia of MOTION, tends to continue moving forward at the original speed. The result: the upper body lurches forward. In both cases the upper body's inertia resists the change in motion that the bus has imposed on the lower body via floor/handrail contact.

Both situations illustrate inertia of rest (first case) and inertia of motion (second case). Key idea: only the part of the body in contact with the bus is accelerated/decelerated; the rest catches up only via the spine.

SHORT_ANSWER · EASY · 12-NIOS-PHY-LM-02

Q3. A 200-kg stone is moving toward you at 2 m/s. At the same time, a 2-kg bullet at 100 m/s is also moving toward you. Explain with reasons which one is more dangerous and requires more force to stop.

Answer: Momentum of stone: $p_{\text{stone}} = 200 \times 2 = 400 \text{ kg}\cdot\text{m/s}$. **Momentum of bullet:** $p_{\text{bullet}} = 2 \times 100 = 200 \text{ kg}\cdot\text{m/s}$. The STONE has higher momentum (400 vs 200 kg·m/s), so a greater impulse is needed to bring it to rest — i.e. either a larger force or a longer interaction time. By $F \cdot \Delta t = \Delta p$, the average force to stop each depends on the time Δt over which the stopping happens. The stone, being slow, can usually be stopped over a longer time (you can step aside, brace, or use a wall); the bullet, being fast, gives an extremely short interaction time, so even though its momentum is smaller, the FORCE required to stop it is enormous (~10⁵ N in a millisecond) and its penetration is dangerous. Both are dangerous, but in different ways: the stone delivers a larger total impulse if it hits; the bullet delivers a much larger force per unit area (due to small contact area and short Δt). For 'requires more force to stop' over a typical short stopping time, the bullet usually wins.

Compare momenta ($p = mv$) and then discuss the role of stopping time in determining force. Both are dangerous; the bullet has lower momentum but higher force-per-unit-time.

SHORT_ANSWER · MEDIUM · 12-NIOS-PHY-LM-03

Q4. Apply Newton's third law of motion to explain the movements of a ball during a game of football.

Answer: Examples of action-reaction during a football game: (1) Kicking the ball: the player's foot exerts a force on the ball (action). The ball exerts an equal and opposite force on the foot (reaction) — that's why a hard kick can hurt. (2) Running on the ground: the foot pushes the ground backward (action); the ground pushes the foot (and player) forward (reaction). Without this reaction, the player couldn't accelerate. (3) Heading the ball: the head pushes the ball forward (action); the ball pushes the head backward (reaction) — which is why headers must be timed carefully. (4) Ball bouncing off the goalpost: the ball exerts a force on the post; the post exerts an equal and opposite force on the ball, sending it bouncing back. (5) Goalkeeper diving and grabbing the ball: the keeper's hands exert a force on the ball; the ball exerts a force on the hands (which the keeper feels as a sting). In every case, the action and reaction act on DIFFERENT bodies (foot vs ball, ground vs foot, etc.) so the two forces do not cancel.

Look for at least three examples of action-reaction pairs and the key point that the two forces act on different bodies.

SHORT_ANSWER · EASY · 12-NIOS-PHY-LM-04

Q5. Newton's First Law (Law of Inertia) states that 'a body continues to be in a state of rest or of uniform motion in a straight line unless it is acted upon by a net external force.' Now roll a ball along the ground and observe its motion. Write your observation and explain why the ball comes to rest after travelling some distance even though no external force seems to act on it.

Answer: **Observation:** a ball rolled along the ground slows down and eventually stops, even though no obvious push or pull is applied after it leaves the hand. **Explanation:** there ARE external forces on the ball — they're just not obvious. Two main ones: (1) Friction between the ball and the ground — a contact force opposing the motion. Even smooth surfaces have some friction; rough surfaces (grass, carpet, gravel) have a lot. (2) Air resistance — pushes back on the ball as it moves through the air. Negligible for a heavy ball at low speeds, significant for light or fast objects. Newton's first law is NOT violated. The 'unless acted upon by a net external force' clause is doing all the work here — the ball does eventually come to rest precisely because friction and air resistance ARE external forces acting on it. If you rolled a ball on a perfectly smooth, frictionless surface in vacuum, it would continue forever — confirming the law.

The misconception is that 'no force acts'. In fact, friction and air resistance are external forces. The first law works perfectly — the ball slows because there IS a force, just not an obvious push.

SHORT_ANSWER · EASY · 12-NIOS-PHY-LM-02

Q6. Observe your surroundings and give TWO examples of friction that are useful and TWO examples of friction that are not useful. Explain how you would increase friction where it is useful and reduce friction where it is not.

Answer: USEFUL friction: (1) Walking — friction between the soles of shoes and the ground allows us to walk without slipping. Increase by using shoes with grooved/rubber soles; reduce by walking on smooth wet tiles. (2) Writing with a pen on paper — friction between the pen tip and the paper transfers ink. Increase by using rougher paper; reduce by using extremely glossy paper (then the pen slips). **NOT-USEFUL friction:** (1) Engine parts rubbing against each other — friction wears the metal and wastes energy as heat. Reduce by using lubricants (engine oil), bearings (ball bearings), or by polishing the surfaces. (2) Air resistance on a car or aeroplane — slows it down and wastes fuel. Reduce by streamlining the body shape (aerodynamics), smoothing the surface, retracting wheels in flight. **Methods to increase friction:** roughen the surface, use rubber/grip materials, increase the pressing force (heavier load increases normal force). **Methods to reduce friction:** use lubricants (oil, grease, graphite), use ball or roller bearings, polish/smooth the surfaces, use compressed air between the surfaces.

Look for two useful and two non-useful examples, plus methods to increase (rough surface, rubber, more weight) and decrease (lubricant, bearings, smooth surface) friction.

SHORT_ANSWER · EASY · 12-NIOS-PHY-LM-07

Q7. Which diagram represents balanced forces? How do opposing forces affect the motion of an object if the forces are balanced?

Answer: Balanced forces: when two or more forces acting on an object are equal in magnitude and act in exactly opposite directions, so that their vector sum is zero. Pictorially, two equal-length arrows pointing in opposite directions along the same line of action represent balanced forces. **Effect on motion when forces are balanced:** Net force = 0. By Newton's second law, $F = ma$, so $a = 0$. This means: (i) if the object is at rest, it remains at rest; (ii) if the object is moving, it continues to move at constant velocity (same speed, same direction). Balanced forces do not change the state of motion of the object. They may, however, change the SHAPE of a deformable object (e.g. squeezing a sponge from both sides) — but they do not change the body's translational motion.

Balanced forces $\Rightarrow$ zero net force $\Rightarrow$ zero acceleration $\Rightarrow$ no change in motion (rest stays rest, motion stays at constant velocity).

SHORT_ANSWER · EASY · 12-NIOS-PHY-LM-02

Q8. Take a solid object from your surroundings and measure its mass. Lift the load with the help of a rope at a constant acceleration to the roof of your house. Measure the height of the roof. Let the load cover this height in 1 minute. Calculate the tension in the rope.

Answer: Sample values (substitute your measured ones). Mass of object $m = 5$ kg. Height $h = 3$ m. Time taken $t = 60$ s. Take $g = 10$ m/s². Acceleration: assuming the object starts from rest and is lifted with constant acceleration, $h = (1/2) a t^2 \Rightarrow a = 2h / t^2 = 6 / 3600 = 0.00167$ m/s² (very small because 1 minute to lift 3 m is slow). Free-body diagram of the load: weight mg pulling down (50 N), tension T pulling up. Newton's second law (upward positive): $T - mg = ma \Rightarrow T = m(g + a) = 5 \times (10 + 0.00167) \approx 50.008$ N. The tension is only marginally greater than the weight because the acceleration is very small. If the object were lifted much faster (a comparable to g), the tension would be roughly $2mg$; if it were lowered with downward acceleration g , the tension would drop to zero (free fall).

Apply Newton's second law: $T - mg = ma$. Tension exceeds weight when accelerating upward. Compute a from $h = \frac{1}{2} a t^2$.

LONG_ANSWER · MEDIUM · 12-NIOS-PHY-LM-08

Q9. Suppose you are standing on a weighing scale in a lift. If the lift is going DOWN with a downward acceleration of 9 m/s^2 , what would be the reading of the weighing scale? Justify your answer with explanation. (Take $g = 10 \text{ m/s}^2$ and your mass = 60 kg .)

Answer: Let your mass $m = 60 \text{ kg}$, downward acceleration $a = 9 \text{ m/s}^2$, take downward as positive. Forces on you: weight mg downward ($= 600 \text{ N}$), normal force N upward from the scale. Newton's second law (downward positive): $m g - N = m a \Rightarrow N = m (g - a) = 60 \times (10 - 9) = 60 \text{ N}$. The weighing scale reads what it pushes you with (Newton's third law) — i.e. the normal force N . So the scale would read 60 N , which corresponds to an apparent mass of $60 / 10 = 6 \text{ kg}$. You feel almost weightless! If the lift were in FREE FALL ($a = g$), N would be zero — true weightlessness. This is the principle of zero-gravity training in aircraft flying parabolic dives.

Apparent weight = $m(g - a)$ when descending with acceleration a . The lift accelerates down faster than gravity alone, so you barely press on the scale.

SHORT_ANSWER · MEDIUM · 12-NIOS-PHY-LM-03

Q10. It is generally observed that roads on hilly areas or mountain roads are winding upwards rather than going straight up. Give reasons.

Answer: Winding mountain roads reduce the EFFECTIVE GRADIENT (slope) of the road. A road that climbs height h over a winding length L has gradient h/L ; a road that climbs the same height over a straight-up length L' has gradient h/L' . Since $L > L'$ (winding is longer), the winding gradient is gentler. Reasons why a gentler gradient is preferred: (1) Less force needed against gravity — $F = mg \sin \theta$. With a smaller angle θ , the gravitational opposing force is smaller, so vehicles can climb with smaller engine power. (2) Better friction — on a steep slope, the normal force $N = mg \cos \theta$ is small ($\cos \theta$ is small for steep θ), so friction (μN) is reduced. On a gentle slope, friction can supply enough traction to drive up without slipping. (3) Safer braking on descent — a steeper slope means brakes have more weight component working against them; gentler slopes are easier to control. (4) Vehicles' maximum slope capability is limited (often around $6-7^\circ$). Direct routes up a mountain often exceed this; winding switchbacks bring the gradient within capability. So winding roads trade extra distance for the ability to climb (and safely descend) the mountain.

Key concepts: gradient = $\sin \theta$; required force to climb = $mg \sin \theta$; reducing θ requires longer (winding) path. Friction ($\mu mg \cos \theta$) is larger on gentler slopes.

SHORT_ANSWER · MEDIUM · 12-NIOS-PHY-LM-08